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Reference Architecture Specification for Drone Systems

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***Abstract*— Unmanned Aerial Systems (UASs) are useful for performing missions ranging from simple to tedious/dangerous ones. In recent years, UAS architectures have been introduced by different groups with different viewpoints. However, these architectures do not take into account all UAS features and their interactions. Thus, the task of developing a new UAS following one of the existing architectures is very difficult. Therefore, to ease the UAS development task, in this paper, we provide a reference architecture specification for future drone systems. It includes the different blocks (functions) of the UAS and their interactions. The blocks are divided into four main groups: flight navigation, flight control, flight management, and mission management.**

Keywords*— *Drones, Unmanned Aerial Systems (UASs), Blocks, Reference Architecture, Drone System Development.

I. Introduction

Drones (i.e., aircrafts without human on-board) fly autonomously or remotely controlled by human operators [1]. The drones suit missions that are very dirty, dangerous, or tedious for humans. To perform a mission, an unmanned aerial system (UAS[1]) needs to be developed [2]. The UAS includes an air segment (i.e., the drones), a ground segment, and a communication means. The ground segment controls and supervise the drones. The communication means supports the UAVs interactions with the ground segment, and external services (e.g., maps service, validating a trajectory, etc.) [3].

In the recent years, the development of drone systems has gained a lot of interest. To ease the development of the drone systems, a number of architectures has been introduced (e.g., Paparazzi [5], ICAROUS [4], Open JHU APL [6], CSAT [8], USAL [7], FACE [10] and MIT/LL [9]. But, the architectures have been proposed with a particular interest without taken the overall drone system functions into account. For example, the USAL and Paparazzi architectures have only focused on the drone control, while the flight cooperation and planning are considered by JHU APL, ICAROUS, and CAST. In addition, the obstacle detection and avoidance capability has been taken into account by the MIT/LL architecture, while the FACE architecture interest is on the drone's safety and security. In addition, the existing architectures do not specify explicitly interactions between the different system functions. Therefore, the development of drone systems that require high safety and autonomy levels is very difficult.

In the context of the COMP4DRONES project [11], we face the existing architectures' limitations by providing a reference architecture for drone systems. To do so, we identified the different building blocks of the UAV systems and interactions between them (more details about the blocks and their interactions can be found in [11]). Based on the identified building blocks and their interactions patterns, we have clustered the different blocks into four main groups: flight control, flight management, flight guidance, and mission management. These groups are then linked together to form the reference architecture of the future drone systems.

In this paper, we also identified the requirements/needs underlying the reference architecture. These requirements are divided into two main groups: architecture requirements and requirements for architecture elements. Furthermore, we present how the different architecture blocks can be clustered and instantiated based on the drone autonomy level (e.g., Remotely Piloted Aircraft (RPA) and semi-autonomous drone).

The remaining of this paper is structured as follows. Section II reviews the existing drone architectures and identifies their limitations. The requirements underlying the reference architecture are presented in Section III. A general structure for the drone system and its building blocks is presented in Section IV. The reference architecture is provided in Section V. Finally, the conclusions and the future work are described in Section VI.

II. Related Work

Recently, a number of UAS architectures has been introduced to support/enable the development of drone systems. The architectures include Paparazzi [5], ICAROUS [4], Open JHU USAL [7], APL [6], MIT/LL [9], CSAT [8], and FACE [10]. In following, we present and analyse briefly the existing architectures.

Following the service-oriented architecture (SOA), the ICAROUS architecture has been proposed [4]. It includes services for providing various drone functional blocks such as plan a path, geo-fence preservation, detection and avoidance of obstacles, plan a task, etc. The services are used for developing complex drone applications. Also, the architecture provides a framework for conflicts'- monitoring and resolving. Furthermore, to achieve a safe operation, the conflict resolvers/monitors are managed through a decision-making software.

For maximizing the drone software system's flexibility, the Paparazzi architecture is introduced from the start as a distributed architecture [5]. It has been developed for enabling the simultaneous' operations of multi drones and ground stations. In the Paparazzi architecture, there is a server that includes each drone's configuration (i.e., flight plan, settings, etc.), and extra data (e.g., weather, topography, maps, etc.). The drone messages are also processed via the server agent and stored to be analysed later.

[1] In this paper, we use the terms drone and UAV to refer to the UAS air segment. We also use drone system, UAV system, and UAS as a reference to the air and ground segments and their interactions.

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To achieve a cooperative behaviour between drones for accomplishing surveillance missions [6], Johns Hopkins University Applied Physics Laboratory (JHU APL) has proposed an autonomous UAV and a cooperative drones' architecture. The JHU APL architecture includes two control layers provided by two modules (i.e. low-level and high-level mission controllers). It also contains a wireless communication layer for facilitating the communication with other drones, and human controllers at the ground control stations (GCSs), and payloads. The JHU APL architecture enables the operation of tiny UAVs, and is considered as a first-generation architecture (introduced in early 2000s). The architecture is not using SOA concepts because of the simple and relatively few UAV payloads.

To integrate sensing, control, and navigation capabilities of drones into a single drone system, an open architecture has been proposed [7]. The architecture is a service-based with a UAV Service Abstraction Layer (USAL). The USAL layer supports the control of the avionics and flight system through a set of high-level services. In the open architecture, each major function/block of the system is encapsulated through a service container. It then becomes a module of the open drone architecture.

UAVs for performing Co-operative Search, Acquisition, and Track (CSAT) missions have been introduced by MIT and Aurora Flight Sciences [8]. The main goal is to handle the co-operative track and search functions by good algorithms. Also, a high-level architecture of drone systems is proposed. The architecture includes a module to plan a flight and to control the drone. The module contains three main blocks: a vision block, a planning block, and an auto-pilot block. The cooperative behavior among the UAVs is enabled via the information sharing.

An architecture for drone's data and software services is proposed by MIT Lincoln Laboratory (MIT/LL) [9]. A main feature of the developed service-based architecture is the ability to accommodate detect and avoid functionality. The detect/avoid function is executed by leveraging ground-based surveillance components. In addition, as a long-term objective, drone-based detection and avoidance capability will be accommodated into the architecture for preventing collisions with other drones. Furthermore, the architecture services are clustered into two groups based on the real-time requirements of the services.

To have an open avionics environment suitable for all drones' types, Future Airborne Capability Environment (FACE) [10] consortium has been established. The consortium defined a reference architecture that is developed by a set of logical elements: input/output, portable functions, operating system, and platform-specific and transport services. The architecture also introduced standardized interfaces to enable communication among the different logical elements. Furthermore, the architecture has three profiles: security, safety, and general purpose.

The different UAS architectures discussed above have considered different blocks/functions of the unmanned aerial system: (1) the USAL and Paparazzi architectures focus on the drone's control; (2) the ICAROUS architecture puts much focus on the flight planning sub-system; (3) the CAST and JHU APL architectures enable the drones' co-operation; (4) MIT/LL architecture is interested in the development of detection and avoidance block; (5) the FACE architecture is more oriented towards the safety and security aspects of the UAV. In addition, the existing architectures have specified the blocks at high level of abstraction without taking their functional interactions into account. Thus, to support the development of future drone system and increase their safety and autonomy level, there is a need for a comprehensive *reference architecture* for drone systems that includes the overall drones' building blocks (i.e., taking into account the different system aspects).

III. Requirements for Drone System Architecture

To specify the reference architecture, there is a need to identify/collect its underlying requirements. The primary stakeholders for the drone architecture are the engineers and developers that will implement the architecture in their applications. Thus, most of the effort in designing the architecture should be to meet the engineers'/developers' needs for the drone system. The needs can be divided into two groups: architecture requirements and requirements for architecture elements (described in the next sub-sections).

A. *Overall Architecture Requirements*

The following are generic requirements that have been collected from researchers and industrials to help defining exactly what a modular/customizable architecture for drone systems needs to specify [13]:

Generic Req. 1. The architecture shall support the specification of system functions independently from their implementation (i.e., independent from any specific programming language).

Generic Req. 2. The architecture shall allow and facilitate the mapping to platform dependent models (i.e., annotation, extension of SW languages, components, HW processors, and other resources the system functions will be mapped to).

Generic Req. 3. The architecture shall enable the capture of the communication behavior among modules/functions.

Generic Req. 4. The architecture shall enable low coupling between the system modules.

Generic Req. 5. The architecture shall provide mechanisms to explicitly describe overall drone system and its configuration, and moreover to facilitate the instantiation of modules.

Generic Req. 6. The architecture shall support and facilitate use of libraries for enabling reuse of already developed modules, and modules of same type shall be compatible.

Generic Req. 7. The architecture shall support homogeneous encapsulation (common I/O interfaces) of same type of algorithms for seamlessly pick and replace.

B. *Specific Requirements*

In the following, we present requirements more related to specific architecture's elements. The specific requirements are presented and associated to the general functionalities and sub-functionalities. Such functions is directly related to and supports the different activities that need to be performed since a request is received to do a mission, until the completion of its successful execution. The activities include supporting activities (i.e., perception, actuation, communication, and storage), flight operation's activities (i.e., flight planning, flight guidance, and flight control), flight management's

activities (i.e., health management, payload management, and data management), and mission management's activities (i.e., mission planning and supervision).

(1) Requirements for Supporting Activities

The supporting activities are required to enable the execution of the core functions of the drone system. They include: perception, actuation, communication, and storage.

Specific Req. 1. The **perception** shall provide geo-referenced position, attitude, velocity upon a global time scale (e.g., GPS and UTC), weather information, and the status of the UAV environment.

Specific Req. 2. The **actuation** shall (a) compute the command for each actuator from the command vector; and (b) prevent, if possible, saturation of actuators that would reduce the stability of the flight (priorities between attitude, heading, and position can be defined).

Specific Req. 3. The **position and velocity** control shall (a) calculate the desired attitude and thrust while considering constraints on speed and supporting a speed control input when used; and (b) support the control and constrain of the required speed metrics, i.e., airspeed (fixed-wing) or ground-speed (rotorcraft).

Specific Req. 4. The **attitude and rate control** shall calculate the command vector to stabilize the orientation of the UAV while taking into account constraints on rotation speeds and to stabilize the body rates of the UAV when used with rates inputs.

Specific Req. 5. The **communications** shall support interactions between system parts through (a) data links for command and control and the UAV payload data; (b) a simultaneous uplink and downlink capability; and (c) radio links for digital message transmission.

Specific Req. 6. The **data storage** shall (a) provide a means of inputting/storing data; (b) be able to communicate with a server to receive, extract, and push data; and (c) be capable of including additional storage without the need to have a major hardware re-configuration to meet the growth of storage requirements.

(2) Requirements for Flight Operations' Activities

To perform/operate a flight, it needs to be planned and executed safely by different flight controllers. The following are the requirements of the flight operation's functionalities.

Specific Req. 1. The **trajectory planning** shall (a) be able to find a sequence of valid waypoints that moves the UAV from a source to a destination based on the route planned by the mission planner; (b) be capable to perform fuel calculations, weight and balance take-off and landing data calculations, and terrain avoidance warning and minimum altitude calculations; and (c) be able to re-plan the trajectory when needed.

Specific Req. 2. The **trajectory validation** shall (a) evaluate the validity of the flight plans requested and provide alternative flight plans when possible if the original flight plan requested was denied; (b) be able to receive the authorization request for an alternative flight plan proposed and perform the complete authorization process; (c) modify airspace allocation when a flight manager or operation manager becomes unavailable; and (d) detect trajectory conflicts between different agents.

Specific Req. 3. The **trajectory execution** shall coordinate execution of the drone trajectory (i.e., perform waypoint navigation), calculate the set-points for position, velocity, and attitude controllers while considering the physical constraints and properties of the drone.

Specific Req. 4. The **detection and avoidance** shall (a) obtain and process data about the UAV surroundings; (b) detect an obstacle from the UAV environment's map; (c) calculate a valid new trajectory to avoid the obstacle; and (d) be configurable, to select the sensor or algorithm used in the sensor data processing and map generation.

Specific Req. 5. The **geo-fence preservation** shall (a) monitor the drone position and the distance between the drone and the geo-fence, and generate a warning if the distance becomes less than a certain threshold; (b) generate commands and execute them to fly away the drone from the fence; and (d) alter the drone trajectory if required when actions taken to preserve the geo-fence.

Specific Req. 6. The **commander executor** shall (a) receive the commands from mission supervision or trajectory execution and execute the commands in a safe manner; and (b) allow the pilot to safely take-off, land, and navigate the drone manually.

(3) Requirements for Flight Management's Activities

During the flight operation, different aspects need to be managed such as power, fault, payload, and data to keep the flight safe. In the following, we list the requirements posed by these management aspects.

Specific Req. 1. The **power management** shall (a) be capable of restoring power in sufficient time to avoid loss of critical mission data or loss of UAV control during power outages; (b) enable continues power supply for executing critical phases of a mission; and (c) provide actual power status and remaining operation time.

Specific Req. 2. The **fault management** shall (a) detect faults during UAV boot-up and mission execution; (b) provide management for faults and create triggers for failsafe actions; and (c) switch to emergency mode if an error condition (fault) is detected.

Specific Req. 3. The **diagnostic** shall generate and store diagnostics data (out of monitored status) for delivery to mission supervision, and to fault and power management.

Specific Req. 4. The **payload management** shall (a) allow the setting of the payload parameters (e.g., exposure time, when to start capture and when to stop capture, and orientation of the payload); (b) coordinate the execution of the payload work plan; (c) capture mission-specific data and execute mission-specific action (e.g., drop a package); and (d) be able to provide processed and unprocessed data to the user/ground controller.

Specific Req. 5. The **flight logger** shall keep track of (a) all flight-related state variables such as position, orientation, and speed; (b) the status of individual platform components such as motors, battery, and processor operation; (c) the status of the sensors and their measurements, as well as the actuators and their values; and (d) faults occurred during systems execution.

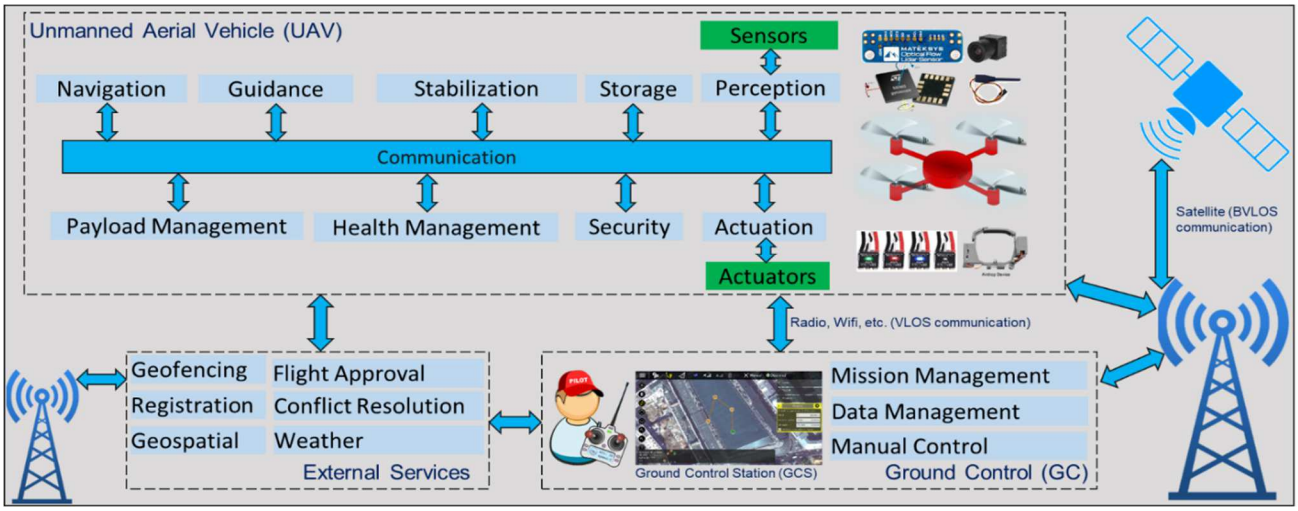

Figure 1: The building blocks of UAS and their interactions with external services

Specific Req. 6. The **prognostics** shall (a) foresee problems related to the battery charge as being insufficient or putting at risk the execution of the mission with a certain safety; (b) foresee situations of possible collision danger based on the planned trajectory, the objects detected, the UAV current location and expected trajectory as can be derived from the UAV sensors; and (c) identify possible hardware failures based on anomalous signals.

Specific Req. 7. The **payload data analytics** shall extract high level information from raw data and perform predictive analysis to forecast the status of the drone environment.

(4) Requirements for Mission Management's Activities

Similar to managing the drone flight, the drone mission should be also planned and supervised.

Specific Req. 1. The **mission planning** shall be capable of (a) importing map information and providing point-and-click route and payload planning; (b) storing mission plans and exporting them; and (c) changing the mission plan during mission execution.

Specific Req. 2. The **mission supervision** shall (a) receive, format, process, store, and retrieve payload and flight data and perform exploitation of payload data; (b) provide dynamic mission and sensor re-planning during the mission execution; (c) provide the operator a caution/warning when the UAV has a malfunction; and (d) have the ability to receive and control payloads.

IV. Building Blocks of Drone Systems

Based on the requirements listed in Section III and to enable the development of drone systems, we have identified the different architecture building blocks of drone systems and their interactions patterns [11]. A building block represents a capability and serves as reference for the creation of its implementation, while an architecture pattern expresses structural organization of the system blocks and relationships between them [12]. In this section, we provide a general structure for drone systems based on the existing architectures. This structure includes the system building blocks and their relationships/interactions. We also describe in brief these blocks and more details can be found in [13].

In general, the building blocks of unmanned aerial systems can be divided into two main clusters based on the system deployment: the drone/UAV and the ground control (GC). These two clusters also interact with the external services as represented in Figure 1.

First, the different blocks of a UAV include guidance, navigation, control, storage, perception, actuation, security, payload management, and health management. These blocks are responsible for executing a mission defined/guided by the ground control. They also interact internally with each other, and communicate externally through Wi-Fi, radio, and satellite-based communications (see Figure 1) [14].

Second, the ground control has a ground control station (GCS) for performing mission and data management. It also includes a manual control of the UAV either by a pilot via a remote control or via the ground control station [15]. Finally, the external services enable the drone operations by providing weather information, maps, trajectory validation, drone registration, etc. It also keeps track of the drone and warns the pilot when the geo-fence area is violated [3]. The above can be clustered into four groups: core functions, drone management, mission management, and support functions. A summary of all the blocks of the reference architecture is shown in Figure 2. In the following sub-section, we describe them briefly.

A. *Core Functions*

The core functions group makes the drone able to fly from a location to another. It contains three building blocks: flight planning, flight guidance, and flight control.

1) ***Flight Planning***

The flight planning refers to determining the desired, and valid path of travel (the "trajectory") from the vehicle's current location to its destination, as well as desired changes in rotation, velocity, and acceleration for following the desired path. It consists of two sub-blocks: trajectory planning and trajectory validation [16]. First, the *trajectory planner* provides a sequence of way-points based on a mission's objectives (e.g., mapping and surveillance), and constraints such as terrain, weather, air space and fuel [17]. Second, the *trajectory validation* is the process of communicating with the external services to validate the flight trajectory. The external services can process drone plan and check for compliance with applicable national regulations.

Figure 2: Overall blocks of a drone system

2) ***Flight Guidance***

The flight guidance refers to the process of monitoring and controlling the movement of a vehicle from one place to another [18]. It consists of four main sub-blocks: trajectory execution, detect and avoid, geofencing, and command executor. First, the *trajectory execution* defines how the UAV should move between the path's way-points [19]. This is an automated function that takes the specifications of the UAV into account to generate a path. The path consists of segments of circles and lines compatible with the drone's turn and climb capabilities.

Second, *detect and Avoid (DAA)* block enables drones integration safely into the civilian airspace, and avoiding collisions with buildings, power lines, other aircraft, birds and other obstacles [20]. This block observes the drone's environment, detects whether a collision is imminent, and generates a new flight path in order to avoid collision. It contains two blocks: detection and avoidance blocks. Third, the *geofencing* block is responsible for keeping the drone flying without violating the geo-fence area defined by the regulation authorities [21]. It contains two functions: geo-awareness and geo-fence preservation. Finally, the *command executor* is responsible for mode switching and executing the different mission commands. The commands can be: navigation commands, do commands, and condition commands.

3) ***Flight Control***

Drone control refers to manipulating the forces, by way of thrusters, steering controls, etc., needed to execute commands while maintaining vehicle stability [22]. This block includes flight control, position and velocity control, and attitude and rate control. First, the *flight control* specifies changes in the UAV's thrust, yaw, pitch, and roll to follow the specified path (typically represented as a sequence of line and circle segments) through the position and altitude controls. Second, the *position and velocity* controller takes the inputs from the trajectory generation and compute the desired attitude and desired thrust. Finally, the *attitude and rate* controller takes the desired attitude (or body rates), and compute the desired command to keep the drone stable, most of time around body axis (roll, pitch, yaw).

B. Drone Management

To make the drone able to fly safely and in an efficient manner, a number of functions are required for managing the drone health, payload, and data (see Figure 2). In the following, we describe these functions.

1) ***Health Management***

The health management block is responsible for analyzing flight data and for responding to problems in the power supply or faults that can occur in the drone engine, sensors, actuators, etc. It consists of three main blocks: power management, fault management, and diagnostics. First, *power management* continuously monitors important power supply parameters, while dealing with the varying power demands of the many aspects of the UAV's operation [23]. It also optimizes the usage of the power source. Second, *fault management* is in charge of detecting and accommodating failures of the UAV [24]. Faults in UAV are commonly represented as either actuator, plant, or sensor failures. The fault management technique depends on the type of the failure experienced. Finally, *diagnostics* enables the detection of problems' occurrences. It is responsible for real-time monitoring of key performance indicators of a drone, collecting and processing health data with intelligence, so that it can diagnose the real cause of a problem, and then alerting on abnormal conditions and deviations.

2) ***Payload Management***

The payload management block manages the UAV payload to execute certain mission [25]. It consists of three blocks: payload coordinator, payload controller, and data acquisition. First, *payload coordinator* takes the mission plan as an input and keeps track of the UAV position. When a location is reached where data acquisition (e.g., capture an image) or an action needs to be taken by the drone (e.g., drop a package), specific commands are issued to the payload controller or the data acquisition block to perform the drone mission. Second, *payload controller* executes the commands coming from the payload coordinator. Such actions are either performed by sensors (e.g., change a camera angle), or through actuators (e.g., pick up a package). Finally, *data acquisition* is the process to measure real world physical conditions needed in the drone mission through signals

sampling (e.g., status of a collapsed building), and converting the captured samples into digital values (e.g., constructing 3D image of the building) that can be later on manipulated by a computer.

3) *Data Management Block*

The data management block consists of capabilities for data persistence and storage, data distribution, semantic transformation, etc. These functions can be used in an online streaming mode in which the data are processed as they are received to enable real-time analytics. First, the *prognostics* block serves as a predictive analytics engine of the drone system. It relies on historical data of drone operation and performance, engineering and physics properties of the drone, and modeling information. Second, the *flight logger* keeps track of the UAV status, and records it for latter analysis by the prognostics block. By default, logging is started automatically when drone is arming, and stopped at disarming phase. A new log file is made for each drone mission. Third, the *storage* block is responsible for storing all the needed information to execute a mission. Example information needed for executing a flight includes flight plans, geo-fence limitation, UAV model (e.g., UAV power supply). During mission execution, it also stores the generated information about the vehicle health during the flight (e.g., power level, engine temperature, etc.), and information about the drone environment. Finally, the *payload data analytics* encapsulates a set of functions for data modeling, analytics and other advanced data processing, such as rule engines.

C. *Mission Management Block*

The mission management block is used for planning and monitoring the execution of a drone mission (i.e., mission planning and mission supervision) [26]. First, *mission planning* is the process of producing a detailed flight schedule which concerns the calculation of the route for UAV(s) and the work plan of the payloads (sensors and actuators). A correct mission plan should meet the mission constraints set for the mission objective. Second, *mission supervision* represents a set of functions that the UAS exposes through user interface to enable human interactions with the UAS. It allows drone operators to connect with and control a UAV and its payloads, either through setting parameters for autonomous operation or via allowing direct control of the UAV, usually from the mission start until landing or the end of the mission.

D. *Support Functions*

The *support functions* are cross cutting functions that enable other UAS blocks execution. They include actuation, perception, and communication (see Figure 2).

1) *Actuation*

The actuation is the functionality for writing data and control signals to an actuator to execute the flight commands. It contains two elements: mixer and actuators drivers. The ***mixer*** receives commands (e.g., turn left) and convert them into motor commands and ensuring that some drone limits are not exceeded. The *drivers* are a set of software modules that have the actual interface with the physical actuators to perform the specific actions.

2) *Perception*

The perception block includes the necessary functions for providing the information required by the other architecture blocks [27]. It includes: UAV status (i.e., attitude, heading, positioning, and velocity), UAV surrounding, weather, and sensor's drivers. First, an *attitude* block provides attitude and heading information for the drone including roll, pitch and yaw. These measurements can then be used to derive an estimate of the drone attitude. Second, the *drone position* can be estimated using Global Navigation Satellite System (GNSS-based) or, when GNSS are denied using methods such as Simultaneous Localization and Mapping (SLAM). Third, the *velocity estimator* provides the drone speed. The speed can be directly measured using a speed sensor, or estimated using other sensors such as the drone's camera. Fourth, the *UAV surrounding* provides information about the drone environment including static and moving obstacles, and weather information such as wind speed. Finally, *drivers* are software modules that have a direct interface with the physical sensors to acquire the needed data.

3) *Communication*

Communication connects sensors, actuators, controllers, gateways and other edge systems [28]. The communication mechanisms take different forms, such as a bus (local to an underlying system platform or remote), or networked architecture for command and control and service data exchange with external systems. Also, Quality of Service (QoS) characteristics such as latency, bandwidth, jitter, reliability, redundancy and resilience should be taken into account. The communication can have four forms. First, a *data link* is the way of connecting the UAV with the external infrastructure for the purpose of sending and receiving digital information (i.e., service data communication). Second, *telemetry* is a digital two-way data stream dedicated to ground control, which can both send data about the flight down to a ground control station, and send command up to the autopilot. Third, *internal communication* is used for exchanging data between the different blocks of the UAV or the ground control station. Such communication can be performed using publish-subscribe model or through directed links between the blocks. Fourth, *communication security* is the prevention of unauthorized users from accessing communications medium, while still exchanging data between the UAV system blocks.

V. THE REFERENCE ARCHITECTURE

In this section, we present how the different blocks can be clustered and instantiated based on the drone autonomy level (e.g., Remotely Piloted Aircraft (RPA) and semi-autonomous drone) to identify the architecture topology. Then, the overall architecture is specified based on the identified clusters.

A. *Remotely Piloted Aircraft System (RPAS)*

ICAO explains the term Remotely Piloted Aircraft (RPA) [31] as "An unmanned aircraft that is controlled from a remote station expected to be integrated with the air traffic management system similar to manned aircraft and, the real-time control is provided through a licensed remote pilot." ICAO also defined the term RPA system as "A number of configurable elements including a remotely-piloted aircraft, its associated remote ground station(s), the required command/control links and any other system elements as required, at any time during flight operation."

Following the definition of RPAS and the building blocks described in Section IV, the distribution of the blocks to the drone platform and the ground station with a (licensed) pilot are shown in Figure 3. Most of the building blocks/functions are performed from the ground, while only the automatic control is done through the drone platform following the commands received from the ground.

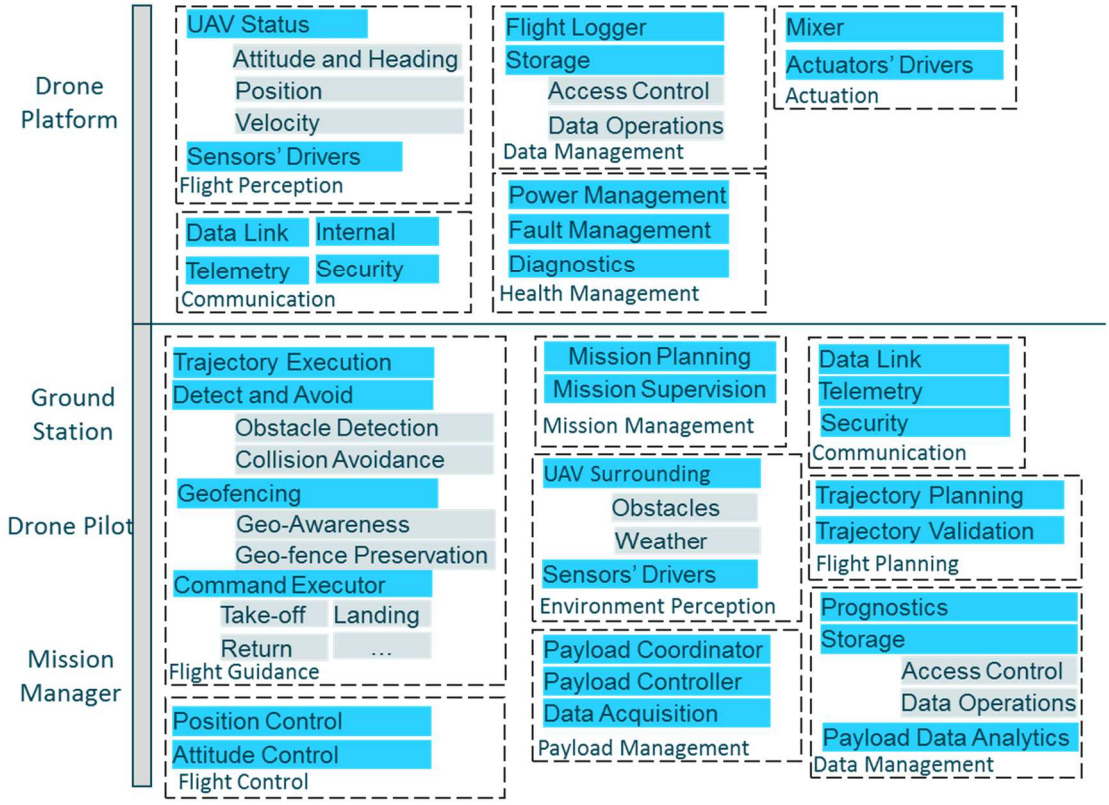

Figure 3: Distribution of the building blocks for RPAS drone

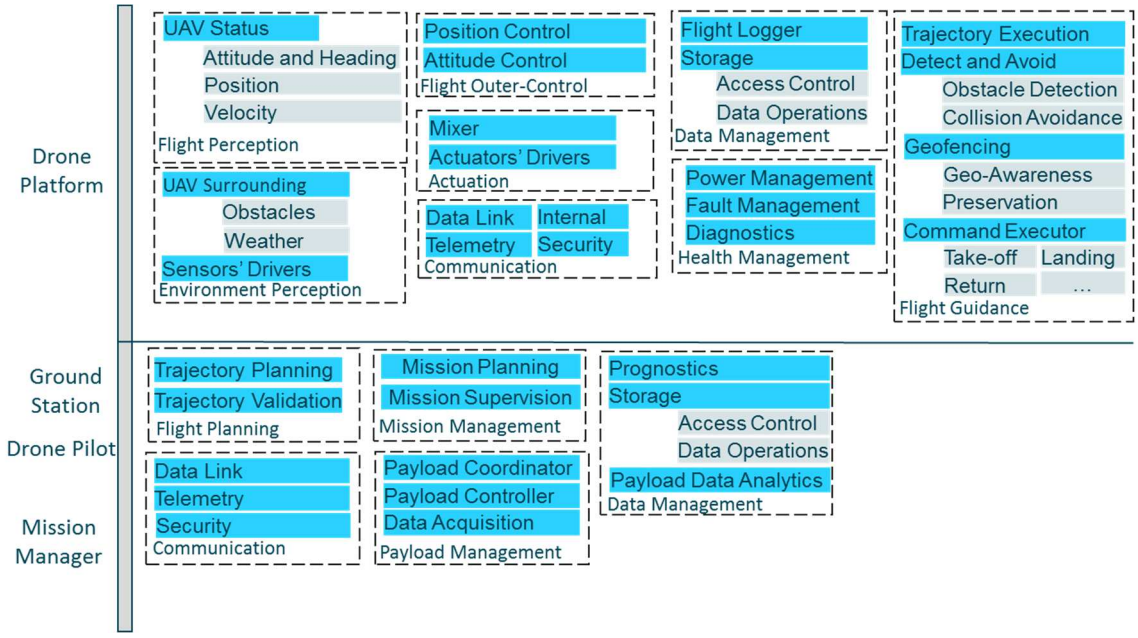

Figure 4: Distribution of the building blocks for semi-autonomous drone

B. Semi-Autonomous Drone (BVLOS)

A (semi-)autonomous drone is a drone that can operate with little human intervention [32]. In other words, the drone can take off, carry out missions, and land in an autonomous manner. In the case of such type of drones, a management software co-ordinates missions and controls the aircraft instead of a human pilot (i.e., the drone is piloted by software instead of a human). To achieve such drone autonomy, most of the UAS functions need to be automated and put on the drone platform to reduce the human interactions as shown in Figure 4.

C. Overall Architecture

The building blocks can exist either in the drone platform or in the ground control based on the drone autonomy level as described in the previous sections. Therefore, the architecture topology can be defined through clustering the building blocks based on their dependences (coupling). The main clusters identified are: flight control, flight management, flight navigation, mission management, and communication as shown in Figure 5.

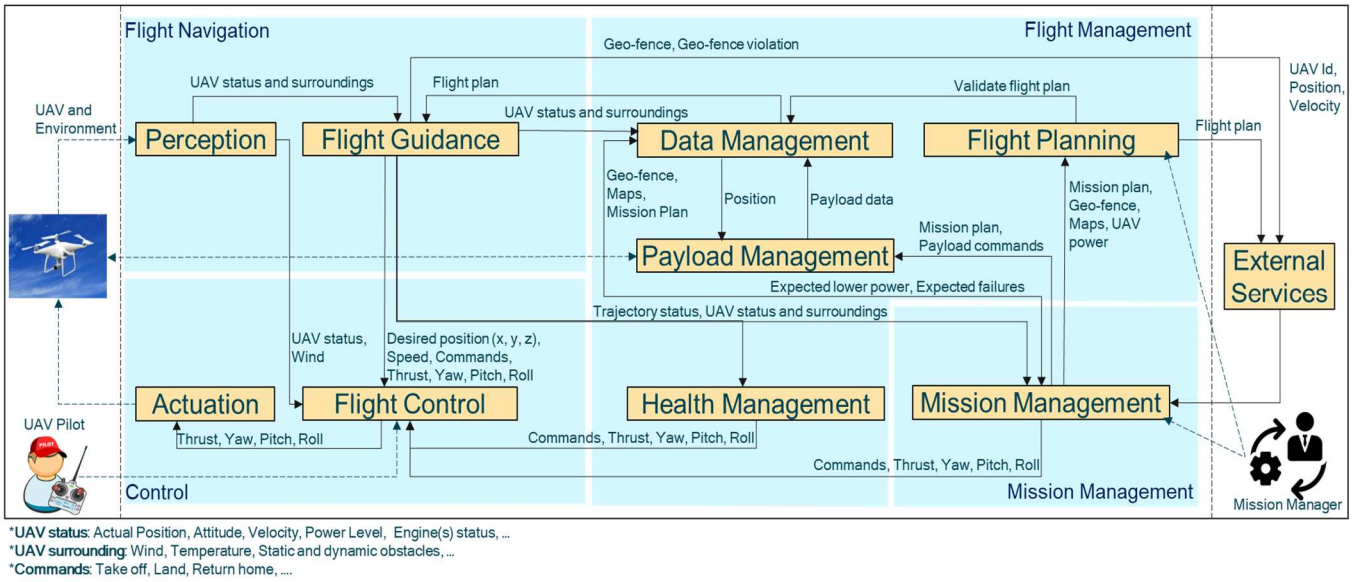

Figure 5: The overall drone system architecture

The overall reference architecture of a drone system is shown in Figure 5. This architecture shows the interactions between the different blocks of the UAS. It is divided into four main groups: flight navigation, flight control, flight management, and mission management. First, the flight navigation includes the drone perception to gather information needed to navigate the drone from one location to another, while avoiding obstacles and preserving the geo-fence using the flight guidance. Second, the flight control executes the guidance commands to fly the drone from one place to another through the drone actuation. It also executes commands coming from the pilot directly.

Third, the flight management contains functions for planning the flight trajectory and managing the UAV payload, data, and health. Fourth, the mission management has the mission planning and supervision functionality that are managed by the mission manager. Finally, there are external services that provide information for mission and flight planners to plan a valid mission and trajectory.

VI. Conclusions and Future Work

In this paper, to face the existing architectures' limitations and to ease the development of a drone system, we provided a reference architecture for drone systems. To do so, the different building blocks of the UAV systems and interactions between them are identified. Then, based on the identified building blocks and their interactions patterns, we clustered the different blocks into four main groups: flight control, flight management, flight guidance, and mission management. Later on, these groups are linked together to form the reference architecture. In this paper, we also presented the requirements underlying the reference architecture. These requirements are divided into two main groups: architecture requirements and requirements for architecture elements. Furthermore, we present how the different architecture blocks can be clustered and instantiated based on the drone autonomy level (e.g., Remotely Piloted Aircraft and semi-autonomous drone).

In the future, we are planning to (a) identify more blocks and patterns, (b) formally document the blocks and patterns; (c) apply the concepts in this paper to real scenarios from the use-cases of the project; and (d) provide an implementation for the reference architecture.

COMP4DRONES Project

The COMP4DRONS project received a fund from ECSEL Joint Undertaking (JU) with a grant agreement No 826610. The JU receives the support from European Union's Horizon 2020 research and innovation programme and France, Spain, Austria, Czech Republic, Belgium, Italy, Latvia, Netherlands. The total budget of the project is 28,590,748.75 EUR (excluding ESIF partners). The project has been running since 1st October 2019.

As shown in Figure 6, the COMP4DRONES partners are covering important geographical area in Europe to ensure that project outcomes will have a large impact on European Drones Technology Industry.

Figure 6: Partners of the COMP4DRONES project

COMP4DRONES project is focusing on developing an ecosystem which is realized as an integrated multi-vendor and modular drone architecture solution with a corresponding tool support.

Challenges (both research and development) addressed in the project cover concerns related to developing safe embedded systems for UAVs.

There are four technical objectives to support the general goal of the project that allows the development of drone ecosystem. The objectives are: Modular Architecture for Drones, Safe and Smart Navigation, Trusted Communication, Tools for System Design and Verification.

Within the project, there are five use cases to cover a wide spectrum of drone applications. The use cases are a strong aspect of COMP4DRONES project. They are segmented applications for autonomous drones into different areas: transport, logistics, construction, agriculture as well as surveillance and inspection.

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